

Phish360 Dataset EDA

Note: This notebook is part of the Anti-Phishing Benchmark Assessment Tool.

Libraries used in this file:

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2. Legitimate Class Data Exploration 2.1 Reading Legit parquet file 2.2 Missing Data 2.3 Average Text Length 2.4 Legit Brand Distribution 2.5 Detected Text Languages 2.6 SSL Distribution 2.7 Top 10 TLD & Domains 2.8 URL Length Stats 2.9 Top 10 FLD & Subdomains

1. Phish samples

```
import pandas as pd
```

1.1 Reading Phish parquet file

```
%%time
# Replace 'path_to_file' with your local path
phish_samples = pd.read_parquet('path/to/Phish360_phish.parquet')
```

```
CPU times: total: 1.27 s
Wall time: 856 ms
```

```
phish_samples.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 4332 entries, 4917 to 10747
Data columns (total 22 columns):
```

#	Column	Non-Null	Count	Dtype
0	dataset_name	4332	non-null	object
1	folder_name	4332	non-null	object
2	Class	4332	non-null	object
3	brand	4332	non-null	object
4	URL	4332	non-null	object
5	TLD	4215	non-null	object
6	Domain	4215	non-null	object
7	FLD	4215	non-null	object
8	Subdomain	2098	non-null	object
9	SSL	4332	non-null	bool
10	image_path	4332	non-null	object
11	translated_tf_text	4319	non-null	object
12	trafilatura_text	4319	non-null	object
13	trafilatura_text_language	4312	non-null	object
14	translated_bs_text	4323	non-null	object

15	BeautifulSoup_text	4323	non-null	object
16	BeautifulSoup_text_language	4321	non-null	object
17	html2text_text	4327	non-null	object
18	lxml_text	4328	non-null	object
19	html_extract_text	4323	non-null	object
20	full_html	4332	non-null	object
21	html_text_language	4332	non-null	object

dtypes: bool(1), object(21)

memory usage: 748.8+ KB

phish_samples.describe()

	dataset_name	folder_name	Class	brand	\
count	4332	4332	4332	4332	
unique	1	4332	1	47	
top	Phish360	P10001	phish	phish	
freq	4332	1	4332	1272	

		URL	TLD
Domain	\		
count		4332	4215
4215			
unique		4257	289
3189			
top	http://partners.leadfusion.com/leadfusion/well...		com
weebly			
freq		5	2242
144			

	FLD	Subdomain	SSL	...	\
count	4215	2098	4332	...	
unique	3199	1242	2	...	
top	weebly.com	www	False	...	
freq	144	588	2484	...	

		trafilatura_text	\
count		4319	
unique		3675	
top	Alibaba.com\nEnglish\n??\n??\nEspañol\nPortugu...		
freq		36	

		trafilatura_text_language	\
count		4312	
unique		30	
top		English	
freq		2990	

		translated_bs_text	\
count		4323	
unique		3995	

```

top      \n\n\nAlibaba Manufacturer Directory - Supplie...
freq                                           14

BeautifulSoup_text \
count                                           4323
unique                                          4002
top      \n\n\nAlibaba Manufacturer Directory - Supplie...
freq                                           14

BeautifulSoup_text_language \
count                                           4321
unique                                           28
top      English
freq                                           3041

html2text_text \
count                                           4327
unique                                          4073
top      \n\n  *[@]:  at \n  *[:  dot \n  *[*W-L-D...
freq                                           16

lxml_text \
count                                           4328
unique                                          4055
top      \nSign In\n\n\n\t\n          \n\n\t\t\t \n ...
freq                                           9

html_extract_text \
count                                           4323
unique                                          3934
top      Alibaba Manufacturer Directory - Suppliers, Ma...
freq                                           15

full_html
html_text_language
count                                           4332
4332
unique                                          4187
17
top      <html><head>\n<title>Sign In</title>\n<meta n...
English
freq                                           9
4206

[4 rows x 22 columns]

# show the first 10 samples in the dataframe
phish_samples.head(10)

```

	dataset_name	folder_name	Class	brand	\
4917	Phish360	P10001	phish	santander	
4918	Phish360	P10002	phish	sparkasse	
4919	Phish360	P10003	phish	itau	
4920	Phish360	P10005	phish	orange	
4921	Phish360	P10006	phish	xfinity	
4922	Phish360	P10007	phish	impots	
4923	Phish360	P10011	phish	orange	
4924	Phish360	P10013	phish	irs	
4925	Phish360	P10014	phish	irs	
4926	Phish360	P10015	phish	boa	
					URL
					TLD
4917	https://www.ryml-postalsupport.com/banks/retai...				com
4918	https://sparkasse-musterstadt.if-einblick.de/p...				de
4919	http://itoken-att.info/\n				info
4920	https://fluggersshout.com/app/03dca8f1f9990877...				com
4921	http://liquidateam.ga/gi\n				ga
4922	http://crm.carsinia.com/custom/fr/impots_final...				com
4923	https://orangecom1.yolasite.com/\n				yolasite.com
4924	http://irs.gov-validate-details.mitreacris-tra...				ro
4925	https://uberbet188.com/wp-includes/ID3/HRBLOCK...				com
4926	https://agbanimarego.weebly.com/\n				com
					Domain
					FLD
Subdomain	\				
4917	ryml-postalsupport	ryml-postalsupport.com			
www					
4918	if-einblick	if-einblick.de			sparkasse-
musterstadt					
4919	itoken-att	itoken-att.info			
None					
4920	fluggersshout	fluggersshout.com			
None					
4921	liquidateam	liquidateam.ga			
None					
4922	carsinia	carsinia.com			
crm					
4923	orangecom1	orangecom1.yolasite.com			
orangecom1					

4924	mitreacris-trans	mitreacris-trans.ro	irs.gov-validate-
	details		
4925	uberbet188	uberbet188.com	
	None		
4926	weebly	weebly.com	
	agbanimarego		

	SSL	...	trafilatura_text \
4917	True	...	Security Advice\nSantander will never contact ...
4918	True	...	SEPA\nJednolite przekazy pieni ne na terenie c...
4919	False	...	Latam Pass MC Inter\nprodutos e serviços\nConh...
4920	True	...	Bandwidth Limit Exceeded\nThe server is tempor...
4921	False	...	Why Stay Signed In?\nWith this option selected...
4922	False	...	Accueil\n>\nAuthentification\n>\nDonnée-person...
4923	True	...	Pour vous identifier\nIndiquez votre compte Or...
4924	False	...	Fill the form with your details\nName\nAddress...
4925	True	...	Home\nGet Unpaid Refund Status\nUnpaid Refund ...
4926	True	...	Bank of America\nBank of America\nSearch\nVeri...

	trafilatura_text_language \
4917	English
4918	Polish
4919	Portuguese
4920	English
4921	English
4922	French
4923	French
4924	English
4925	English
4926	Turkish

	translated_bs_text \
4917	\n\n\nLogin\n\n\n\n\n\n\n\n\n\n\n\n\n\n\nSantander...
4918	Failure transfers\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n...
4919	\n\n\n\n\n\n\n\n\n\n\n\n\n\n\nCartão de Crédito Banco ...
4920	\n509 Bandwidth Limit Exceeded\n\nBandwidth Li...
4921	\n\nSign in to Xfinity\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n...
4922	Individual taxes.gouv.fr\n\n\n\n\n\n\n\n\n\n\n\n\n...
4923	Orange.com\n...
4924	\nIRS - Get refund\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n...
4925	\n\nHome\n\t\t\t...
4926	Bank of America\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n...

	BeautifulSoup_text \
4917	\n\n\nLogin\n\n\n\n\n\n\n\n\n\n\n\n\n\n\nSantander...
4918	\nPrzekazy pieni ne - Jedno konto do pBatno[ci...
4919	\n\n\n\n\n\n\n\n\n\n\n\n\n\n\nCartão de Crédito Banco ...
4920	\n509 Bandwidth Limit Exceeded\n\nBandwidth Li...
4921	\n\nSign in to Xfinity\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n...
4922	\n\n\nParticulier impots.gouv.fr\n\n\n\n\n\n\n\n...

```

4923  \nOrange.com\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n...
4924  \nIRS - Get refund\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n...
4925  \n\n\nHome\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\t\t\t...
4926  \nBank of America\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n...

```

	BeautifulSoup_text_language	\
4917	English	
4918	Polish	
4919	Portuguese	
4920	English	
4921	English	
4922	French	
4923	French	
4924	English	
4925	English	
4926	Turkish	

```

4917   Santander\n\n>>>>\n\n>>>> # Log on to Online...
4918   [ ![Strona startowa - Sparkasse Musterstadt]...
4919   \n\n>>>>\n\n>>>> ## atenção!\n\n>>>>\n\n>>>> ...
4920   \n\n>>>>\n\n>>>> # Bandwidth Limit Exceeded\n...
4921   \n\n>>>>\n\n>>>> # Sign in to Xfinity\n\n>>>>...
4922   Accueil > Authentification > Donnée-personne...
4923   \n\n>>>>\n\n>>>> ##### Fi...
4925   [ ![Internal Revenue Service United States D...
4926   \n>>>>\n* [ Bank of America ](/)\n\n>>>>\n*...

```

[illegible]

```

4917 Login\n\nSantander\n\nLog on to Online Banking...
4918 Przekazy pieni|ne - Jedno konto do pBatno[ci n...
4919 Cartão de Crédito | Banco Itaú\n\natenção!\n\n...
4920 509 Bandwidth Limit Exceeded\n\nBandwidth Limi...
4921 Sign in to Xfinity\n\nSign in to Xfinity\n\nAd...
4922 Particulier | impots.gouv.fr\n\nAccueil > Auth...
4923 Orange.com\n\nPour vous identifier\n\nIndiquez...

```

```

4924 IRS - Get refund\n\nFill the form with your de...
4925 Home\n\nHome\nGet Unpaid Refund Status\nUnpaid...
4926 Bank of America\n\nBank of America\n\nBank of ...

```

```
full_html
```

```
html_text_language
```

```

4917 <html lang="en-GB"><head><script>var Aes={};A...
English

```

```

4918 <html class="no-touch js" lang="pl"><head>\n<...
English

```

```

4919 <html class="" lang="pt-BR"><head>\n    <!-- ...
English

```

```

4920 <html><head>\n<title>509 Bandwidth Limit Exce...
English

```

```

4921 <html class="light da da-expandable da-expand...
English

```

```

4922 <html lang="fr"><head>\n    <link rel="shortc...
English

```

```

4923 <html lang="fr"><head><meta content="" name="...
English

```

```

4924 <html><head>\n\t<title>IRS - Get refund</titl...
English

```

```

4925 <html><head>\n<meta http-equiv="Content-Type"...
English

```

```

4926 <html lang="en"><head><script type="text/java...
English

```

```
[10 rows x 22 columns]
```

```
# show the last 10 samples in the dataframe
```

```
phish_samples.tail(10)
```

	dataset_name	folder_name	Class	brand	\
10738	Phish360	P02520	phish	phish	
10739	Phish360	P02521	phish	phish	
10740	Phish360	P02523	phish	phish	
10741	Phish360	P02524	phish	phish	
10742	Phish360	P02525	phish	phish	
10743	Phish360	P02526	phish	phish	
10744	Phish360	P02529	phish	phish	
10745	Phish360	P02530	phish	phish	
10746	Phish360	P02532	phish	phish	
10747	Phish360	P02533	phish	phish	

```
URL
```

```
TLD
```

\	URL	TLD
10738	https://d50b7b5f-6b9f-4132-b1e2-9ce54d890108.i...	id.repl.co

10739	http://onlinesecure-auth1v.ddns.net/BECU/login...	ddns.net
-------	---	----------

10740	http://invoice.bomberos-melilla.com/\n	com
10741	https://discountcope.com/foldas/index.htm\n	com
10742	http://demo.bradescocontrol.vertitecnologia.co...	com.br
10743	https://6d7cd743-f8b9-4146-9ef2-6517f0b26e51.i...	id.repl.co
10744	https://loginontwtter.yolasite.com/\n	yolasite.com
10745	http://roganclinic.com/rfclaw/se365\n	com
10746	https://www.na-citiprepaidcard.com/\n	com
10747	http://arbredenoe.com/access/efs/servlet/efs/l...	com

	Domain	\
10738	d50b7b5f-6b9f-4132-b1e2-9ce54d890108	
10739	onlinesecure-authlv	
10740	bomberos-melilla	
10741	discountcope	
10742	vertitecnologia	
10743	6d7cd743-f8b9-4146-9ef2-6517f0b26e51	
10744	loginontwtter	
10745	roganclinic	
10746	na-citiprepaidcard	
10747	arbredenoe	

	FLD	\
10738	d50b7b5f-6b9f-4132-b1e2-9ce54d890108.id.repl.co	
10739	onlinesecure-authlv.ddns.net	
10740	bomberos-melilla.com	
10741	discountcope.com	
10742	vertitecnologia.com.br	
10743	6d7cd743-f8b9-4146-9ef2-6517f0b26e51.id.repl.co	
10744	loginontwtter.yolasite.com	
10745	roganclinic.com	
10746	na-citiprepaidcard.com	
10747	arbredenoe.com	

	Subdomain	SSL	...	\
10738	d50b7b5f-6b9f-4132-b1e2-9ce54d890108.id	True	...	
10739	onlinesecure-authlv	False	...	
10740	invoice	False	...	
10741	None	True	...	
10742	demo.bradescocontrol	False	...	
10743	6d7cd743-f8b9-4146-9ef2-6517f0b26e51.id	True	...	
10744	loginontwtter	True	...	

10745	None	False	...
10746	www	True	...
10747	None	False	...

trafilatura_text \

10738	La clave personal y todo o cualquier otro meca...
10739	Welcome to BECU Online Banking.Please sign in ...
10740	Gebruik je een publieke of gedeelde computer?Z...
10741	KYC Documentation Update\nDEAR HUIJIE XIAO / M...
10742	Login\nLembrar minha senha\nNão tem acesso?\nR...
10743	1\n2\n3\nValidación de contacto\nValidación fi...
10744	Home\ncontact\nTwitter\nHome\ncontact\nPhone, ...
10745	Microsoft Secure Email\nLogin\nForgot?\nBy cli...
10746	Password Update Notice\nExpress Login. Click h...
10747	Log in to your account\nQuestions?\nFor assist...

trafilatura_text_language \

10738	Spanish; Castilian
10739	English
10740	Dutch; Flemish
10741	English
10742	Portuguese
10743	Spanish; Castilian
10744	English
10745	English
10746	English
10747	English

translated_bs_text \

10738	\n \nRed L...
10739	\n\n Online Banking Login BECU\n\n\n\n\n\n\n\n...
10740	Login My Marktplaats\n\n\n\n\n\n\n\n\n\n\n\n\n\n...
10741	\n\n\n\n\n\nLOAN - BANK in UAE RAKBANK\n\n\n\n\n...
10742	BRADESCO CONTROL\n\n\n\n\n\n\n\n\n\n\n\n\n\n...
10743	Required verification\n\n\n\n\n\n\n\n\n\n\n\n\n\n...
10744	\n\n\n\n\n\n\n\nTwitter\n\n\n\n\n\n\n\n\n\n\n\n\n...
10745	\n\n Microsoft Secure Email Secure File Shar...
10746	\n\nWirecard Northlane\n\n\n\n\n\n\n\n\n\n\n\n\n...
10747	\nUltraBranch Login\n\n\n\n\n\n\n\n\n\n\n\n\n\n...

BeautifulSoup_text \

10738	\n \nRed L...
10739	\n\n Online Banking Login BECU\n\n\n\n\n\n\n\n...
10740	\nLogin Mijn Marktplaats\n\n\n\n\n\n\n\n\n\n\n\n...
10741	\n\n\n\n\n\nLOAN - BANK in UAE RAKBANK\n\n\n\n\n...
10742	\n\n\n\n\nBradesco Control\n\n\n\n\n\n\n\n\n\n\n...
10743	\n\n\n\n\nVerificacion Requerida\n\n\n\n\n\n\n\n...
10744	\n\n\n\n\n\n\n\nTwitter\n\n\n\n\n\n\n\n\n\n\n...
10745	\n\n Microsoft Secure Email Secure File Shar...
10746	\n\nWirecard Northlane\n\n\n\n\n\n\n\n\n\n\n...

```
10747  \nUltraBranch Login\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n\n...
```

	BeautifulSoup_text_language	\
10738	Spanish; Castilian	
10739	English	
10740	Dutch; Flemish	
10741	English	
10742	Portuguese	
10743	Spanish; Castilian	
10744	English	
10745	English	
10746	English	
10747	English	

```

10738 [Cerrar](javascript:; "Cerrar ayuda") \n> ...
10739 [ ![BECU Online Banking - More than just\nmo...
10740 [Marktplaats](https://www.marktplaats.nl/ "M...
10741 \n\n>>>>\n* Personal __\n\n>>>>\n* Accounts\n...
10742 ![Bcontrol\nlogo](/assets/bcontrol_logo-f309...
10743  1 \n\n>>>>\n* [con...
10745 \n:',
sum(phish_samples.full_html.isnull()))
print('the number of missing URLs (phish):',
sum(phish_samples.URL.isnull()))
print('the number of missing images (phish):',
sum(phish_samples.image_path.isnull()))

the number of missing HTML files (phish): 0
the number of missing URLs (phish): 0
the number of missing images (phish): 0

```

1.3 Average Text Length

```

print('the average phish HTML code text length:',
round(phish_samples.full_html.str.len().sum() /
sum(phish_samples.full_html.notnull()), 1))
print('the average phish trafilatura text length:',
round((phish_samples.trafilatura_text.str.len().sum() /
sum(phish_samples.trafilatura_text.notnull()))), 1))
print('the average phish BeautifulSoup text length:',

```

```

round(phish_samples.BeautifulSoup_text.str.len().sum() /
sum(phish_samples.BeautifulSoup_text.notnull()), 1))
# print('the average phish html2text text length:',
round(phish_samples.html2text_text.str.len().sum() /
sum(phish_samples.html2text_text.notnull()), 1))
# print('the average phish html_extract text length:',
round(phish_samples.html_extract_text.str.len().sum() /
sum(phish_samples.html_extract_text.notnull()), 1))
# print('the average phish lxml_text text length:',
round(phish_samples.lxml_text.str.len().sum() /
sum(phish_samples.lxml_text.notnull()), 1))

```

the average phish HTML code text length: 46506.7
the average phish trafilatura text length: 568.8
the average phish BeautifulSoup text length: 1803.6

1.4 Phish Brand Distribution

```
phish_samples.brand.value_counts()
```

phish	1272
facebook	219
paypal	211
chase	182
office	179
boa	130
orange	129
wellsfargo	125
dhl	115
apple	108
alibaba	107
itau	102
yahoo	101
outlook	93
att	90
google	89
microsoft	82
dropbox	82
adobe	82
amazon	77
santander	57
netflix	53
amex	46
bradesco	40
sparkasse	39
sfr	39
nab	38
usaa	37

```
xfinity      35
linkedin     34
cibc         34
bmo          33
natwest      26
aol          26
irs          24
ebay         23
match        22
tmobile      22
caixa        21
mic          19
cox          19
bt           16
impots       15
hsbc         14
navy         13
capitalone   10
allegro       2
Name: brand, dtype: int64
```

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
brands = phish_samples.brand.value_counts()[1:21].index
values = phish_samples.brand.value_counts()[1:21].values
```

```
# Create the plot
```

```
sns.set_style("whitegrid")
plt.figure(figsize=(20,10))
plot = sns.barplot(x=values, y=brands, orient='h')
```

```
# Increase the font size of y-axis labels
```

```
plot.set_yticklabels(plot.get_yticklabels(), fontsize=16)
```

```
# Add labels and title
```

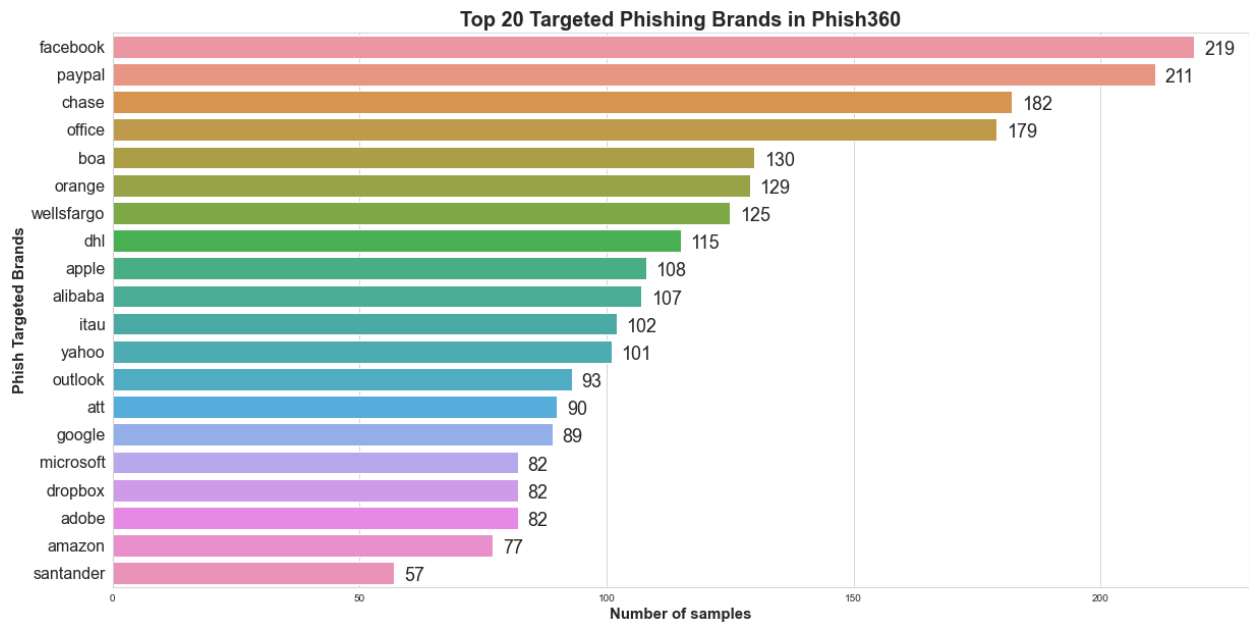
```
plt.xlabel('Number of samples', fontsize=15, fontweight='bold')
plt.ylabel('Phish Targeted Brands', fontsize=15, fontweight='bold')
plt.title("Top 20 Targeted Phishing Brands in Phish360", fontsize=20,
fontweight='bold')
```

```
# Add numbers over the bars
```

```
for p in plot.patches:
    plot.annotate(format(int(p.get_width()), 'd'), (p.get_width()+2,
p.get_y()+0.7), fontsize=18)
```

```
# Show the plot
```

```
plt.show()
```



1.5 Detected Text Languages

```

trafilatura_langs_phish =
phish_samples.trafilatura_text_language.value_counts()

# top 10 languages detected in trafilatura text
trafilatura_langs_phish[:10]

English                2990
French                 297
Portuguese             244
German                 187
Spanish; Castilian     151
Turkish                132
Italian                71
Dutch; Flemish         48
Indonesian             44
Catalan; Valencian     18
Name: trafilatura_text_language, dtype: int64

import matplotlib.pyplot as plt
import seaborn as sns

# langs = trafilatura_langs_phish.index[:30]
# values = trafilatura_langs_phish.values[:30]

langs = trafilatura_langs_phish.index
values = trafilatura_langs_phish.values

# Create the plot
sns.set_style("whitegrid")

```

```

plt.figure(figsize=(10, 10)) # Adjust the figure size accordingly

plot = sns.barplot(x=values, y=langs, orient='h')

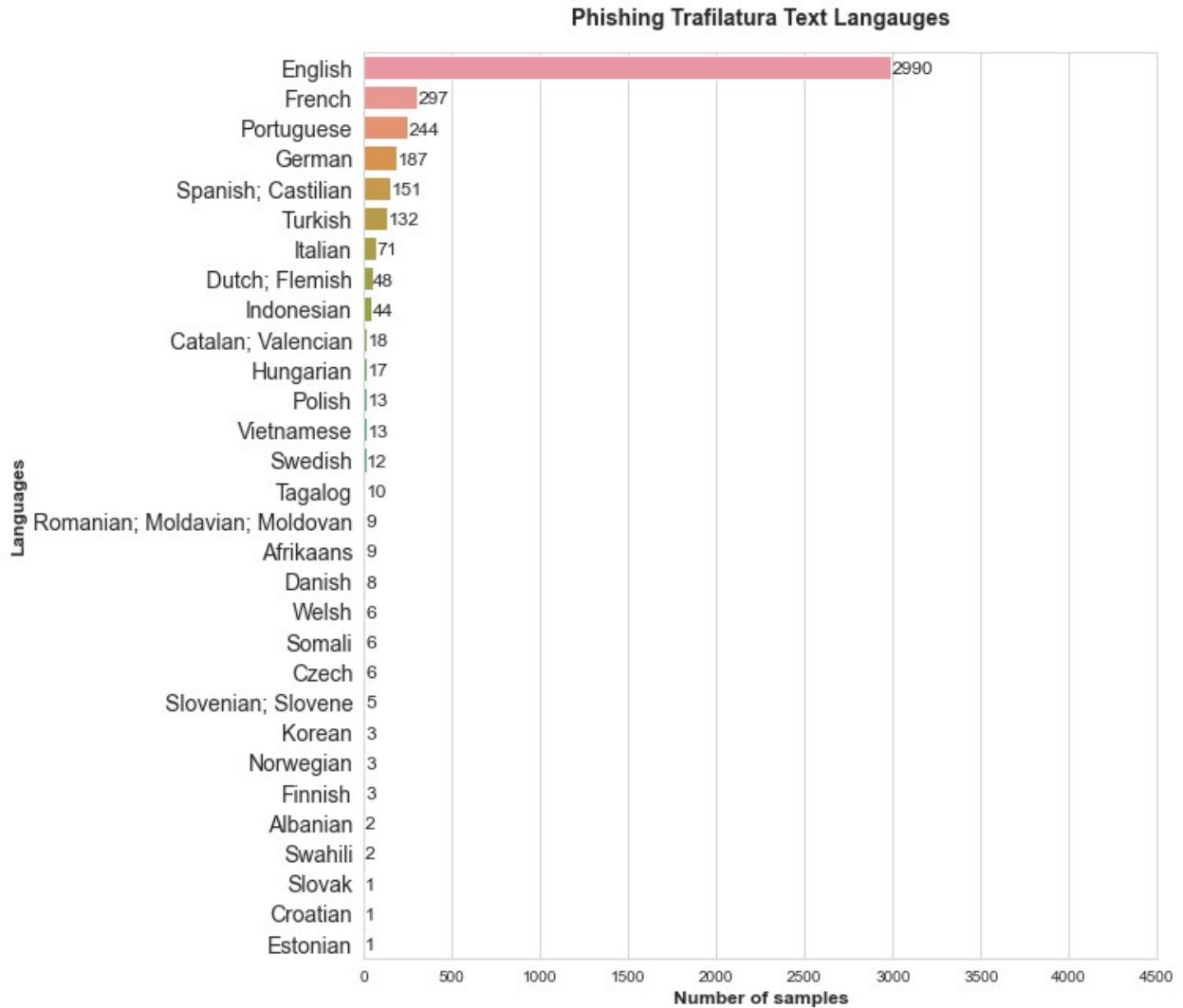
# Increase the font size of y-axis labels
plot.set_yticklabels(plot.get_yticklabels(), fontsize=14)

# Add labels and title
plt.xlabel('Number of samples', fontsize=12, fontweight='bold')
plt.ylabel('Languages', fontsize=12, fontweight='bold')

# plt.title("Phishing samples detected langauges using Trafilatura Text (30 total langauges)")
plt.title("Phishing Trafilatura Text Langauges", fontsize=14,
fontweight='bold', y=1.02)
plt.xlim(0, 4500)
# Add numbers over the bars
for p in plot.patches:
    plot.annotate(format(int(p.get_width()), 'd'), (p.get_width()+5,
p.get_y()+0.6), fontsize=12)
# plot.annotate(format(int(p.get_width()), 'd'), (p.get_width()+2,
p.get_y()+0.7), fontsize=18)
# Adjust the space for labels
plt.subplots_adjust(left=0.2, right=0.9, top=0.9, bottom=0.1) # You
may need to fine-tune these values

# Show the plot
plt.show()

```



```
BeautifulSoup_langs_phish =
phish_samples.BeautifulSoup_text_language.value_counts()
```

```
# top 10 languages detected in BeautifulSoup text
BeautifulSoup_langs_phish[:10]
```

```
English          3041
French           295
Portuguese       252
German           181
Spanish; Castilian 143
Turkish          135
Italian          64
Dutch; Flemish   52
Indonesian       36
Hungarian        14
Name: BeautifulSoup_text_language, dtype: int64
```



```

print('The total number of phishing languages (BS): ',
len(BeautifulSoup_langs_phish))

The total number of phishing languages (BS): 28

langs = BeautifulSoup_langs_phish.index
values = BeautifulSoup_langs_phish.values

# Create the plot
sns.set_style("whitegrid")
plt.figure(figsize=(10, 10)) # Adjust the figure size accordingly

plot = sns.barplot(x=values, y=langs, orient='h')

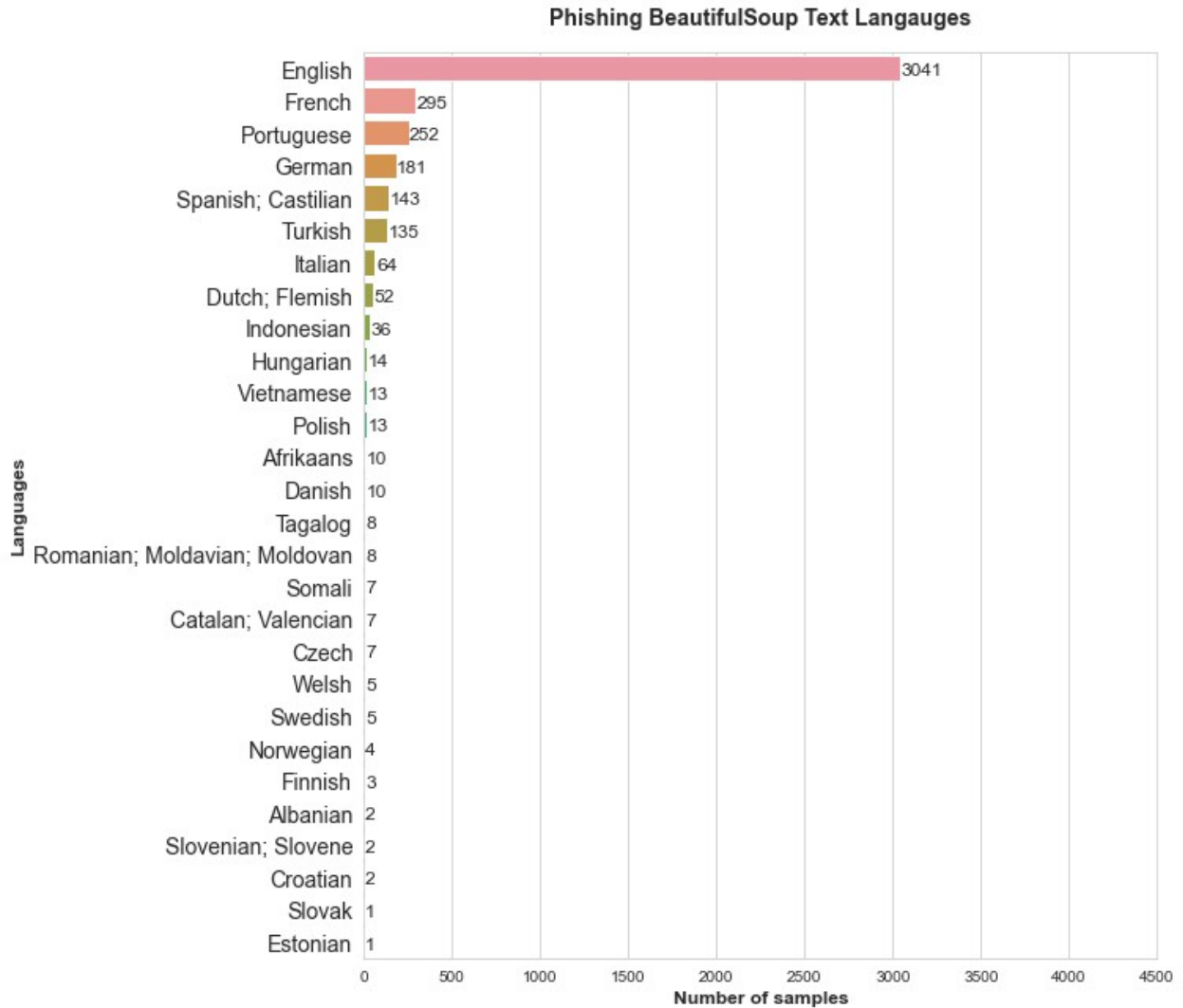
# Increase the font size of y-axis labels
plot.set_yticklabels(plot.get_yticklabels(), fontsize=14)

# Add labels and title
plt.xlabel('Number of samples', fontsize=12, fontweight='bold')
plt.ylabel('Languages', fontsize=12, fontweight='bold')

# plt.title("Phishing samples detected langauges using Trafilatura
Text (30 total langauges)")
plt.title("Phishing BeautifulSoup Text Langauges", fontsize=14,
fontweight='bold', y=1.02)
plt.xlim(0, 4500)
# Add numbers over the bars
for p in plot.patches:
    plot.annotate(format(int(p.get_width()), 'd'), (p.get_width()+5,
p.get_y()+0.6), fontsize=12)
# plot.annotate(format(int(p.get_width()), 'd'), (p.get_width()+2,
p.get_y()+0.7), fontsize=18)
# Adjust the space for labels
plt.subplots_adjust(left=0.2, right=0.9, top=0.9, bottom=0.1) # You
may need to fine-tune these values

# Show the plot
plt.show()

```



```
html_langs_phish = phish_samples.html_text_language.value_counts()
```

```
# top 10 languages detected in html text
```

```
html_langs_phish[:10]
```

```
English          4206
Catalan; Valencian    61
French            21
Croatian          10
Portuguese         6
German             5
Somali             5
Spanish; Castilian  5
Danish             3
Hungarian          2
Name: html_text_language, dtype: int64
```

```

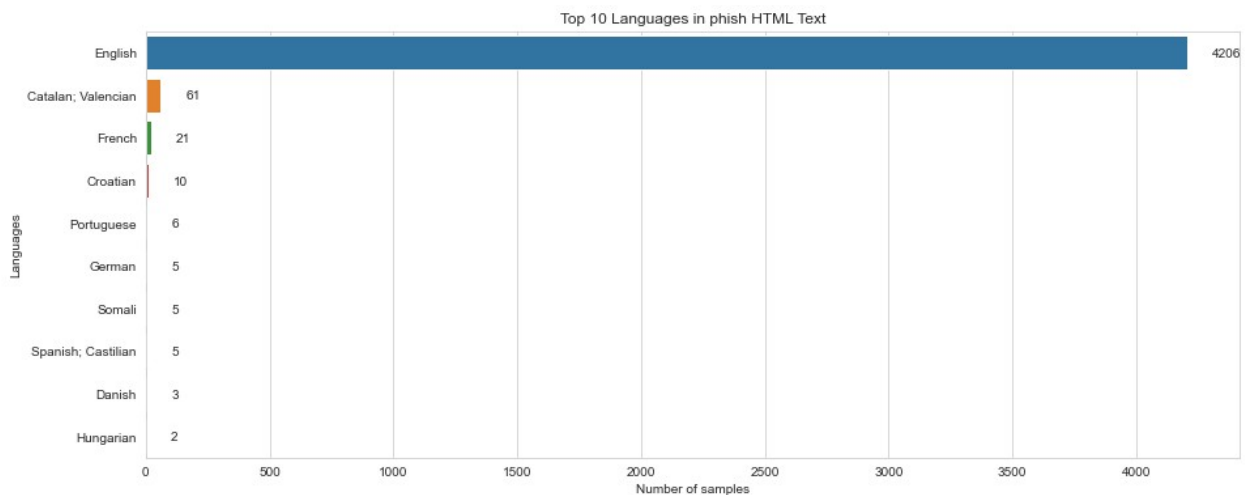
langs = html_langs_phish[:10].index
values = html_langs_phish[:10].values

# Create the plot
sns.set_style("whitegrid")
plt.figure(figsize=(15,6))
plot = sns.barplot(x=values, y=langs, orient='h')

# Add labels and title
plt.xlabel('Number of samples')
plt.ylabel('Languages')
plt.title("Top 10 Languages in phish HTML Text")
# Add numbers over the bars
for p in plot.patches:
    plot.annotate(format(int(p.get_width()), 'd'), (p.get_width()+100,
p.get_y()+0.5))

# Show the plot
plt.show()

```



1.6 SSL Distribution

```

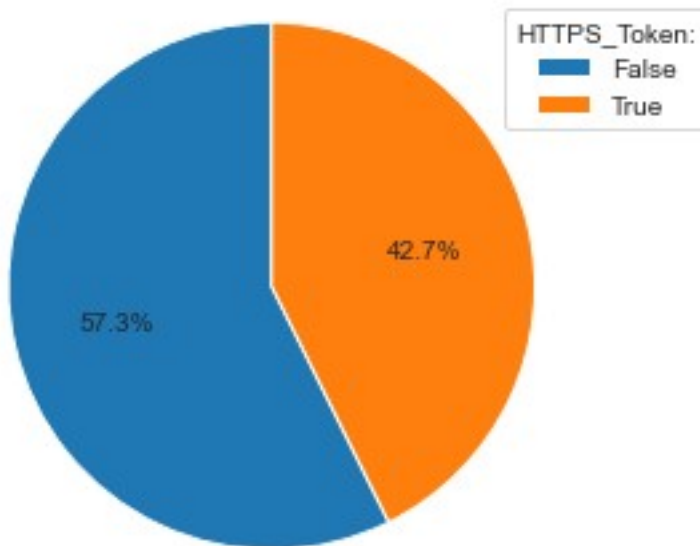
phish_samples.SSL.value_counts()

False    2484
True     1848
Name: SSL, dtype: int64

sizes = phish_samples.SSL.value_counts().values
labels= phish_samples.SSL.value_counts().index
# fig=plt.figure(figsize=(10, 5))
plt.pie(sizes, startangle = 90, autopct = '%1.1f%%')
plt.legend(labels, title = 'HTTPS_Token:')
plt.axis('equal')

```

```
plt.show()
```



1.7 Top 10 TLD & Domains

```
phish_samples.Domain.value_counts()[10]
```

```
weebly      144
000webhostapp  87
webnode     65
google      48
myfreesites 32
webcindario 21
umbler       21
weblium      18
secureserver 17
if-einblick  16
Name: Domain, dtype: int64
```

```
domains = phish_samples.Domain.value_counts()[15].index
counts = phish_samples.Domain.value_counts()[15].values
```

```
# Set the figure size
plt.figure(figsize=(15, 6))
```

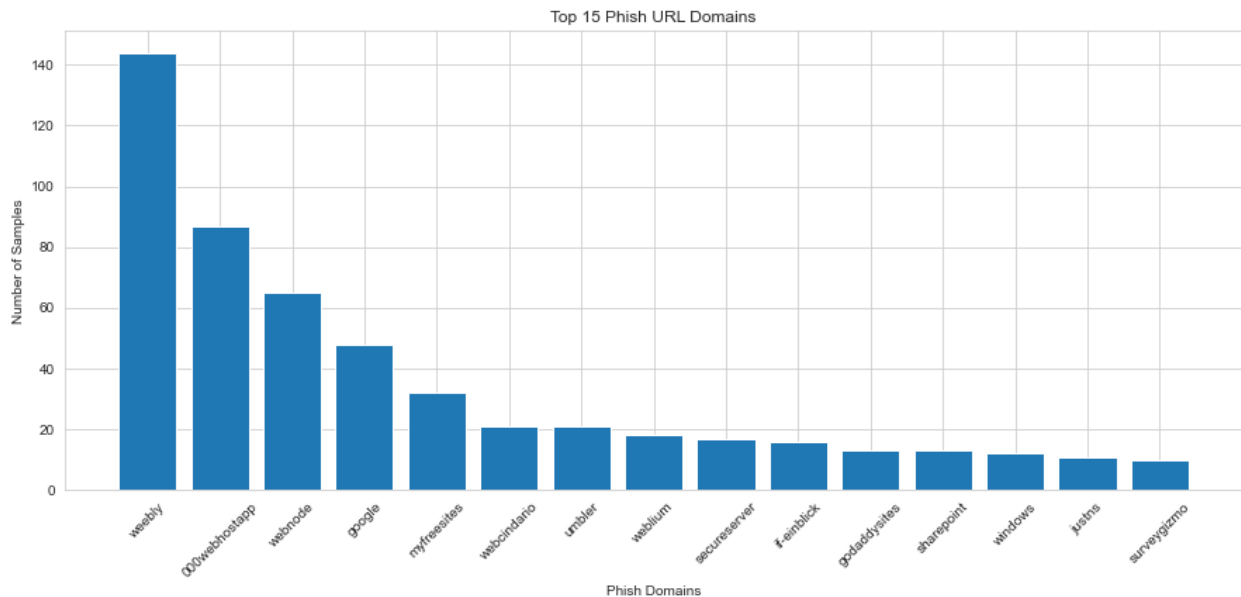
```
# Create the bar plot
plt.bar(domains, counts)
```

```
# Add labels and title
plt.xlabel("Phish Domains")
plt.ylabel("Number of Samples")
```

```
plt.title("Top 15 Phish URL Domains")
```

```
# Rotate the x-tick labels  
plt.xticks(rotation=45)
```

```
# Show the plot  
plt.show()
```



```
phish_samples.TLD.value_counts()[:10]
```

```
com      2242  
net      212  
com.br   113  
org      111  
ru       70  
com.au   64  
de       54  
xyz      52  
info     49  
in       49
```

```
Name: TLD, dtype: int64
```

```
tlds = phish_samples.TLD.value_counts()[:15].index  
counts = phish_samples.TLD.value_counts()[:15].values
```

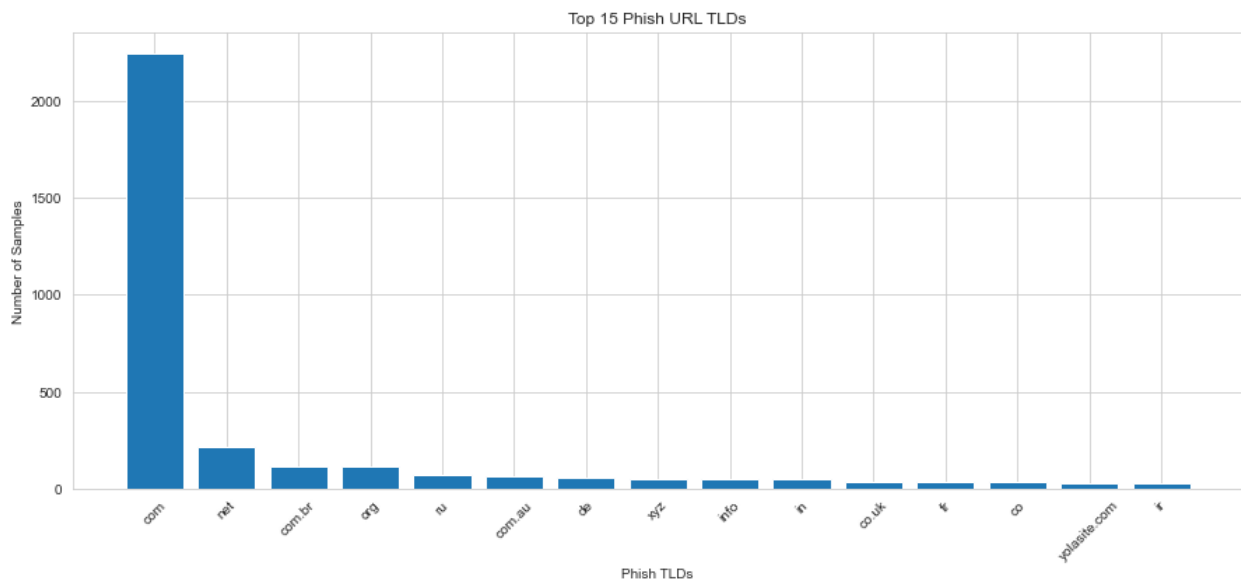
```
# Set the figure size  
plt.figure(figsize=(15, 6))
```

```
# Create the bar plot  
plt.bar(tlds, counts)
```

```
# Add labels and title
plt.xlabel("Phish TLDs")
plt.ylabel("Number of Samples")
plt.title("Top 15 Phish URL TLDs")

# Rotate the x-tick labels
plt.xticks(rotation=45)

# Show the plot
plt.show()
```



1.8 URL Length Stats

```
print('Phish URL average length:',
      round(phish_samples.URL.str.len().sum()/sum(phish_samples.URL.notnull(
      )), 4))
print('Phish URL min length:', phish_samples.URL.str.len().min())
print('Phish URL max length:', phish_samples.URL.str.len().max())
print('Phish URL median length:',
      phish_samples.URL.str.len().median())
```

```
Phish URL average length: 76.1025
Phish URL min length: 19
Phish URL max length: 1642
Phish URL median length: 56.0
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
phish_samples['url_length'] = phish_samples.URL.str.len()
legit_samples['url_length'] = legit_samples.URL.str.len()
```

```
# Create a single subplot for VanNL126k
fig, ax = plt.subplots(figsize=(10, 8))

# Plot phishing dataset
sns.histplot(phish_samples['url_length'], color="red",
label="Phishing", ax=ax)

# Plot legitimate dataset on top of phishing
sns.histplot(legit_samples['url_length'], color="blue",
label="Legitimate", ax=ax)

title_fontsize = 22

# Set the title and axis labels for the subplot
ax.set_title('Phish360 Dataset', fontsize=title_fontsize)
ax.set_xlabel('URL Character Length', fontsize=18)
ax.set_ylabel('Frequency', fontsize=18)

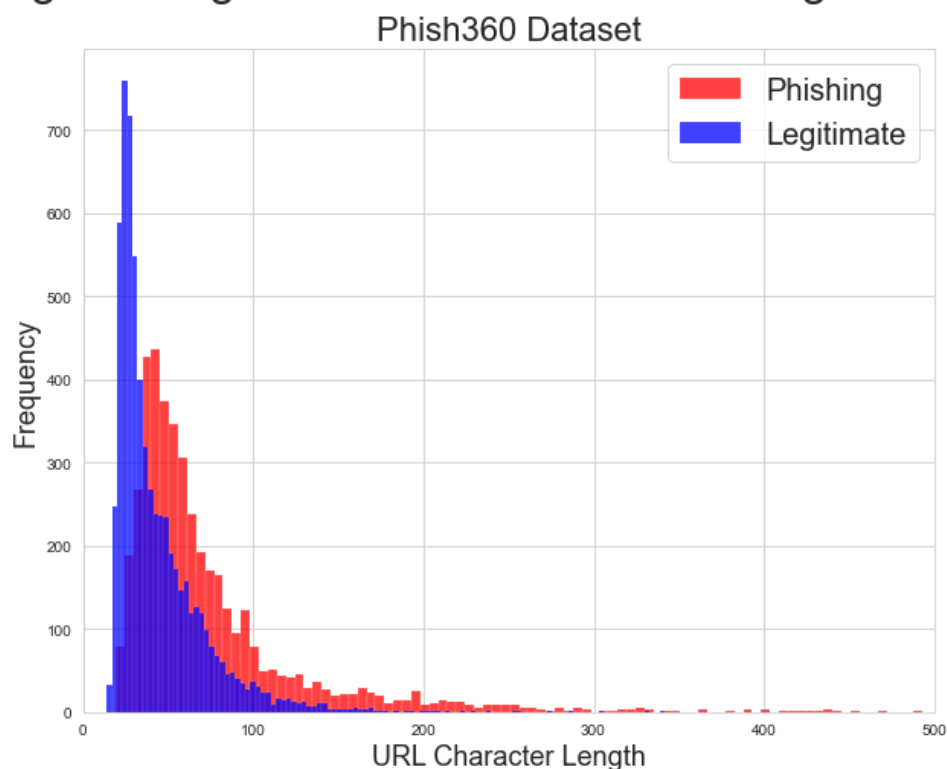
# Set the x-axis limit to 0-800 or adjust as needed
ax.set_xlim(0, 500)

# Add a legend to the subplot
ax.legend(fontsize=20)

# Set the main title
plt.suptitle("Phishing and Legitimate URLs Character Length
Distribution", fontsize=30)

plt.show()
```

Phishing and Legitimate URLs Character Length Distribution



1.9 Top 10 FLD & Subdomains

```
phish_samples.iloc[:, 4:9].head(10)
```

	URL	TLD
\		
4917	https://www.ryml-postalsupport.com/banks/retai...	com
4918	https://sparkasse-musterstadt.if-einblick.de/p...	de
4919	http://itoken-att.info/\n	info
4920	https://fluggersshout.com/app/03dca8f1f9990877...	com
4921	http://liquidateam.ga/gi\n	ga
4922	http://crm.carsinia.com/custom/fr/impots_final...	com
4923	https://orangecom1.yolasite.com/\n	yolasite.com
4924	http://irs.gov-validate-details.mitreacris-tra...	ro
4925	https://uberbet188.com/wp-includes/ID3/HRBLOCK...	com
4926	https://agbanimarego.weebly.com/\n	com

	Domain	FLD
Subdomain		
4917	ryml-postalsupport	ryml-postalsupport.com
www		
4918	if-einblick	if-einblick.de
musterstadt		sparkasse-
4919	itoken-att	itoken-att.info
None		
4920	fluggersshout	fluggersshout.com
None		
4921	liquidateam	liquidateam.ga
None		
4922	carsinia	carsinia.com
crm		
4923	orangecom1	orangecom1.yolasite.com
orangecom1		
4924	mitreacris-trans	mitreacris-trans.ro
details		irs.gov-validate-
4925	uberbet188	uberbet188.com
None		
4926	weebly	weebly.com
agbanimarego		

```
print('the number of unique flds:', len(phish_samples.URL.unique()) -
len(phish_samples.URL))
print('the number of unique flds:', len(legit_samples.URL.unique()) -
len(legit_samples.URL))
```

```
the number of unique flds: -75
the number of unique flds: -38
```

```
print('the number of unique flds:', len(phish_samples.FLD.unique()))
phish_samples.FLD.value_counts()[:20]
```

```
the number of unique flds: 3200
```

weebly.com	144
000webhostapp.com	87
webnode.com	63
google.com	48
myfreesites.net	32
umbler.net	21
webcindario.com	21
weblium.site	18
secureserver.net	17
if-einblick.de	16
godaddysites.com	13
sharepoint.com	13
windows.net	12

```
justns.ru 11
surveygizmo.com 10
angelfire.com 9
office.com 9
securewebsession.com 8
googleusercontent.com 6
creatorlink.net 6
Name: FLD, dtype: int64
```

```
print('the number of unique subdomains:',
len(phish_samples.Subdomain.unique()))
phish_samples.Subdomain.value_counts()[:20]
```

```
the number of unique subdomains: 1243
```

```
www
588
sites
49
mail
27
sparkasse-musterstadt
16
forms
9
secure75
8
shop
6
partners
5
piwik
5
us
4
a2plvc pnl308307.prod.iad2
4
storage
4
secure
4
ecoute-sms-vocal
4
track
4
epl
4
braadesco-com
4
secure.info.updates.chase.log.alertno.id893219803218321073216129312691
```

```

32      4
dropbox.progentrix
4
ib-nab.bankingg.au
3
Name: Subdomain, dtype: int64
phish_samples.iloc[:, 4:9].describe()

```

	URL	TLD
Domain \		
count	4332	4215
unique	4257	289
top	http://partners.leadfusion.com/leadfusion/well...	com
weebly		
freq	5	2242
144		

	FLD	Subdomain
count	4215	2098
unique	3199	1242
top	weebly.com	www
freq	144	588

2. Legit samples

2.1 Reading Legit parquet file

```

%%time
# Replace 'path_to_file' with your local path
legit_samples = pd.read_parquet('path/to/Phish360_phish.parquet')

CPU times: total: 3.36 s
Wall time: 2.38 s

legit_samples.info()

```

```

<class 'pandas.core.frame.DataFrame'>
Int64Index: 6416 entries, 0 to 9475
Data columns (total 20 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   dataset_name          6416 non-null   object
 1   folder_name           6416 non-null   object
 2   Class                 6416 non-null   object
 3   brand                 6416 non-null   object
 4   URL                   6416 non-null   object
 5   TLD                   6408 non-null   object

```

6	Domain	6408	non-null	object
7	FLD	6408	non-null	object
8	Subdomain	5131	non-null	object
9	SSL	6416	non-null	bool
10	image_path	6416	non-null	object
11	trafilatura_text	6398	non-null	object
12	trafilatura_text_language	6395	non-null	object
13	BeautifulSoup_text	6402	non-null	object
14	BeautifulSoup_text_language	6402	non-null	object
15	html2text_text	6407	non-null	object
16	lxml_text	6411	non-null	object
17	html_extract_text	6402	non-null	object
18	full_html	6416	non-null	object
19	html_text_language	6416	non-null	object

dtypes: bool(1), object(19)
memory usage: 1008.8+ KB

legit_samples.head(10)

	dataset_name	folder_name	Class	brand	\
0	Phish360	L01562	legit	unknown	
1	Phish360	L01564	legit	unknown	
2	Phish360	L01565	legit	unknown	
3	Phish360	L01566	legit	unknown	
4	Phish360	L01567	legit	unknown	
5	Phish360	L01568	legit	unknown	
6	Phish360	L01569	legit	unknown	
7	Phish360	L01570	legit	unknown	
8	Phish360	L01571	legit	unknown	
9	Phish360	L01572	legit	unknown	

	URL	TLD
Domain	\	
0	https://www.depicus.com/wake-on-lan/\n	com
depicus		
1	http://liquidsdr.org/doc/tutorial-fec/\n	org
liquidsdr		
2	http://www.csgnetwork.com/ctrcalc.html\n	com
csgnetwork		
3	http://www.dailymusicsheets.com/piano/sheets/\n	com
dailymusicsheets		
4	http://www.mylifetime.com/shows/the-client-list\n	com
mylifetime		
5	http://www.gpsinstallations.co.uk/services/aud...	co.uk
gpsinstallations		
6	http://www.echoecho.com/htmllinks01.htm\n	com
echoecho		
7	https://plato.stanford.edu/entries/voltaire/\n	edu
stanford		
8	http://www.123helpme.com/tps-a-transaction-pro...	com

```
9 https://www.brandsoftheworld.com/logo/save-the... com
brandsoftheworld
```

			image_path	\
0	E:\Datasets\phish360	dataset\trainval\L01562_l...		
1	E:\Datasets\phish360	dataset\trainval\L01564_l...		
2	E:\Datasets\phish360	dataset\trainval\L01565_l...		
3	E:\Datasets\phish360	dataset\trainval\L01566_l...		
4	E:\Datasets\phish360	dataset\trainval\L01567_l...		
5	E:\Datasets\phish360	dataset\trainval\L01568_l...		
6	E:\Datasets\phish360	dataset\trainval\L01569_l...		
7	E:\Datasets\phish360	dataset\trainval\L01570_l...		
8	E:\Datasets\phish360	dataset\trainval\L01571_l...		
9	E:\Datasets\phish360	dataset\trainval\L01572_l...		

```
0 I never believed that this can be functional -...
1 Tutorial: Forward Error Correction\nThis tutor...
2 |\nThis calculator requires the use of Javascr...
3 Piano Sheet Music"... I'm not conscious of the...
4 The Client List Full Episodes, Video & More | ...
5 Here at GPS Installations, we can carry out pu...
6 |\n||\n|\n|\n|\nThe tags used to produce links...
7 Voltaire\nFrançois-Marie dAroutet (1694|1778), b...
8 Transaction Processing Systems(TPS) Definition...
9 Save the Children\nThe logotype in its actual ...
```

	trafilatura_text_language \
0	English
1	English
2	English
3	English
4	English
5	English
6	English
7	English
8	English


```

                                html_extract_text \
0 Welcome to Wake On Lan\n\nndepicus\n\nchanging ...
1 Tutorial: Forward Error Correction - liquidsdr...
2 CSG's Click Through Rate Calculator\n\nHome | ...
3 Daily Music Sheets - Free Downloadable Sheet M...
4 The Client List Full Episodes, Video & More | ...
5 Public Address System Installation - PA System...
6 Links : How to Make a Link - HTML Tutorial\n\n...
7 Voltaire (Stanford Encyclopedia of Philosophy)...
8 Tps - A Transaction Processing System - 1378 W...
9 Save the Children | Brands of the World" | Dow...

```

```

                                full_html
html_text_language
0 <html class="no-js" itemscope="" itemtype="ht...      English
1 <html lang="en-US"><head>\n    <title>Tutoria...      English
2 <html><!-- HTML coding exclusively with the...      English
3 <html xmlns="http://www.w3.org/1999/xhtml" xm...      English
4 <html lang="en"><head>\n    <title>The Client...      English
5 <html class="no-js" lang=""><!--<![endif]--><...      English
6 <html><head>\n<meta http-equiv="pics-label" c...      English
7 <html><!-- <![endif]--><head>\n<meta name="vi...      English
8 <html class="wf-balooopensans-n4-inactive wf-...      English
9 <html dir="ltr" class="js" lang="en"><head><s...      English

```

```
legit_samples.tail(10)
```

	dataset_name	folder_name	Class	brand	\
9466	Phish360	L01552	legit	unknown	
9467	Phish360	L01553	legit	unknown	
9468	Phish360	L01554	legit	unknown	
9469	Phish360	L01555	legit	unknown	
9470	Phish360	L01556	legit	unknown	
9471	Phish360	L01557	legit	unknown	
9472	Phish360	L01558	legit	unknown	
9473	Phish360	L01559	legit	unknown	
9474	Phish360	L01560	legit	unknown	
9475	Phish360	L01561	legit	unknown	

	URL	TLD	\
9466	https://angel.co/provenance\n	co	
9467	http://www.gameitnow.com/Traffic-Light-Games-o...	com	
9468	http://www.graphic-design-employment.com/graph...	com	
9469	http://www.oregon.gov/oha/amh/pd/Pages/approve...	gov	
9470	https://www.vimooz.com/film-festivals/sxsw-sou...	com	
9471	http://www.advertise.com/\n	com	
9472	http://www.publicbookshelf.com/selfhelp/\n	com	
9473	https://www.directlyrics.com/halsey-hopeless-f...	com	
9474	https://www.scribd.com/doc/254693366/Numerical...	com	
9475	https://progressnowcolorado.org/2017/05/11/win...	org	

	Domain	FLD
Subdomain \		
9466	angel	angel.co
None		
9467	gameitnow	gameitnow.com
www		
9468	graphic-design-employment	graphic-design-employment.com
www		
9469	oregon	oregon.gov
www		
9470	vimooz	vimooz.com
www		
9471	advertise	advertise.com
www		
9472	publicbookshelf	publicbookshelf.com
www		
9473	directlyrics	directlyrics.com
www		
9474	scribd	scribd.com
www		
9475	progressnowcolorado	progressnowcolorado.org
None		

	SSL	image_path	\
9466	True	E:\Datasets\phish360 dataset\test\L01552_legit...	
9467	False	E:\Datasets\phish360 dataset\test\L01553_legit...	
9468	False	E:\Datasets\phish360 dataset\test\L01554_legit...	
9469	False	E:\Datasets\phish360 dataset\test\L01555_legit...	
9470	True	E:\Datasets\phish360 dataset\test\L01556_legit...	
9471	False	E:\Datasets\phish360 dataset\test\L01557_legit...	
9472	False	E:\Datasets\phish360 dataset\test\L01558_legit...	
9473	True	E:\Datasets\phish360 dataset\test\L01559_legit...	
9474	True	E:\Datasets\phish360 dataset\test\L01560_legit...	
9475	True	E:\Datasets\phish360 dataset\test\L01561_legit...	

	trafilatura_text	\
9466	We empower brands to make the sourcing and imp...	
9467	Rule as a real traffic controller traffic in t...	

9468 I've been working with graphic design software...
9469 You are here:\n0HA COVID-19 Updates:\nVisit ou...
9470 Filming began in florida on the dramatic horro...
9471 ADVERTISER SOLUTIONS THAT DELIVER REAL RESULTS...
9472 -\nSEX - Avoided Subjects Discussed in Plain E...
9473 Halsey didnt just sing on Episode of Season 44...
9474 Learn more about Scribd Membership\nThis conte...
9475 |\n|\nTHE WINNERS\nRural Colorado\nAfter years...

trafilatura_text_language \

9466	English
9467	English
9468	English
9469	English
9470	English
9471	English
9472	English
9473	English
9474	English
9475	English

BeautifulSoup_text \

9466 Provenance | AngelListStartup JobsRecruitInves...
9467 \nTraffic Light Games Free Online - GameItNow....
9468 Graphic Design Software You Need to Learn for ...
9469 \n\n\n\tOregon Health Authority : Welcome : Pe...
9470 \n\n\n\n\nVIM00Z | Independent Films, Document...
9471 \n\n\n Advertise.com | Online Ad Network | Adver...
9472 \n\n\n\n\nRead Self-help Books Online - Free \n\...
9473 \n\n\n\nHalsey lyrics - Directlyrics\n\n\n\n\n\n\n...
9474 \n\n\nRemoval Notice | Scribd\n\n\n\n\n\n\nSearchSea...
9475 \n\n\n\n\n\nWinners and Losers of the 2017 Color...

BeautifulSoup_text_language \

9466	English
9467	English
9468	English
9469	English
9470	English
9471	English
9472	English
9473	English
9474	English
9475	English

html2text_text \

9466 [](/)\n\n>>>>\n* [Startup Jobs](/jobs)\n\n>>>...
9467 [![Play your favorite game online right\nnow...
9468 \n\n\n>>>>\n\n\n>>>> ## Cookie Control\n\n\n>>>>\n\n...

9469 Skip to main content [![Oregon.gov\nHomepag...
9470 \n\n>>>\n* [Home](https://www.vimooz.com/)\n...
9471 \n> [» *S...
9473 \n\n>>>\n\n>>> # Directlyrics\n\n>>>\n\n>>...
9474 [![Scribd](https://s-f.scribdassets.com/image...
9475 \n\n>>>\n* [Take Action](https://act.beacon....

lxml_text \

9466 Provenance | AngelListStartup JobsRecruitInves...
9467 \n Traffic Light Games Free Online - GameItNo...
9468 Graphic Design Software You Need to Learn for ...
9469 \n \n \n\tOregon Health Authority : ...
9470 \n\n\n\n\n\nVIM00Z | Independent Films, Document...
9471 \n\t\n\t\t\t\tAdvertise.com | Online Ad Network ...
9472 \n \n \n \n\tRead Self-help Books Onl...
9473 \n\n\nHalsey lyrics - Directlyrics\n\n\n\n\n\n\n...
9474 \n \n Removal Notice | Scribd\n ...
9475 \n\t\n\n\t\n\n\t\n\n\t\n\n\t\n\n\tWinners and Lose...

html_extract_text \

9466 Provenance | AngelList\n\n\nStartup Jobs\nRecrui...
9467 Traffic Light Games Free Online - GameItNow.co...
9468 Graphic Design Software You Need to Learn for ...
9469 Oregon Health Authority : Welcome : Peer Deliv...
9470 VIM00Z | Independent Films, Documentary & Film...
9471 Advertise.com | Online Ad Network | Advertiser...
9472 Read Self-help Books Online - Free\n\n\nHome » S...
9473 Halsey lyrics - Directlyrics\n\n\nDirectlyrics\n...
9474 Removal Notice | Scribd\n\n\nSearch Search\nClos...
9475 Winners and Losers of the 2017 Colorado legisl...

full_html

html_text_language

9466 <html prefix="og: http://ogp.me/ns#" lang="en...
English
9467 <html xmlns="http://www.w3.org/1999/xhtml" xm...
English
9468 <html xmlns:og="http://opengraphprotocol.org/...
English
9469 <html class=" js flexbox flexboxlegacy canvas...
English
9470 <html class=" js " lang="en-US"><head>\n<meta...
English
9471 <html class="Firefox Firefox59 cssanimations ...
English
9472 <html xmlns="http://www.w3.org/1999/xhtml" cl...
English

```

9473 <html class="artist" lang="en"><head>\n<meta ...
English
9474 <html xmlns="http://www.w3.org/1999/xhtml" xm...
English
9475 <html class="js" lang="en-US"><head>\n\t<meta...
English

```

```
legit_samples.describe()
```

	dataset_name	folder_name	Class	brand	\
count	6416	6416	6416	6416	
unique	1	6416	1	1	
top	Phish360	L01562	legit	unknown	
freq	6416	1	6416	6416	

		URL	TLD
Domain	\		
count		6416	6408
6408			
unique		6378	196
5692			
top	http://bestreviews.com/best-wireless-routers\n		com
wikipedia			
freq		2	3019
23			

	FLD	Subdomain	SSL	\
count	6408	5131	6416	
unique	5704	467	2	
top	wikipedia.org	www	False	
freq	23	4487	3873	

	image_path	\
count	6416	
unique	6416	
top	E:\Datasets\phish360 dataset\trainval\L01562_l...	
freq	1	

	trafilatura_text	\
count	6398	
unique	6269	
top	Küçük resimler\nBelge _emasl\nEkler\nÖnceki\nS...	
freq	10	

	trafilatura_text_language	\
count	6395	
unique	27	
top	English	
freq	4141	

[illegible]

2.2 Missing Data

```
print('the number of missing HTML files (legit):',
sum(legit_samples.full_html.isnull()))
print('the number of missing URLs (legit):',
sum(legit_samples.URL.isnull()))
print('the number of missing images (legit):',
sum(legit_samples.image_path.isnull()))

the number of missing HTML files (legit): 0
the number of missing URLs (legit): 0
the number of missing images (legit): 0
```

2.3 Average Text Length

```

print('the average legit HTML code text length:',
      round(legit_samples.full_html.str.len().sum() /
            sum(legit_samples.full_html.notnull()), 1))
print('the average legit trafilatatura text length:',
      round((legit_samples.trafilatura_text.str.len().sum() /
            sum(legit_samples.trafilatura_text.notnull()), 1))
print('the average legit BeautifulSoup text length:',
      round(legit_samples.BeautifulSoup_text.str.len().sum() /
            sum(legit_samples.BeautifulSoup_text.notnull()), 1))
# print('the average legit html2text text length:',
#       round(legit_samples.html2text_text.str.len().sum() /
#             sum(legit_samples.html2text_text.notnull()), 1))
# print('the average legit html_extract text length:',
#       round(legit_samples.html_extract_text.str.len().sum() /
#             sum(legit_samples.html_extract_text.notnull()), 1))
# print('the average legit lxml_text text length:',
#       round(legit_samples.lxml_text.str.len().sum() /
#             sum(legit_samples.lxml_text.notnull()), 1))

the average legit HTML code text length: 115180.4
the average legit trafilatatura text length: 4132.5
the average legit BeautifulSoup text length: 9326.6

```

2.4 Legit Brand Distribution

```

#there are no legitimate brand data
legit_samples.brand.value_counts()

unknown      6416
Name: brand, dtype: int64

```

2.5 Detected Text Languages

```

trafilatura_langs_legit =
legit_samples.trafilatura_text_language.value_counts()

# top 10 languages detected in trafilatatura text
trafilatura_langs_legit[:10]

```

English	4141
German	486
Czech	466
Spanish; Castilian	376
Romanian; Moldavian; Moldovan	193
French	148
Turkish	105
Catalan; Valencian	79
Finnish	66

Slovak

55

Name: trafilatura_text_language, dtype: int64

```
print('The total number of legitimate languages (trafilatura): ',  
len(trafilatura_langs_legit))
```

The total number of legitimate languages (trafilatura): 27

```
langs = trafilatura_langs_legit.index  
values = trafilatura_langs_legit.values
```

Create the plot

```
sns.set_style("whitegrid")  
plt.figure(figsize=(10, 10)) # Adjust the figure size accordingly  
plot = sns.barplot(x=values, y=langs, orient='h')
```

Increase the font size of y-axis labels

```
plot.set_yticklabels(plot.get_yticklabels(), fontsize=14)
```

```
plt.xlabel('Number of samples', fontsize=12, fontweight='bold')  
plt.ylabel('Languages', fontsize=12, fontweight='bold')
```

plt.title("Top 10 Languages in Legit Trafilatura Text")

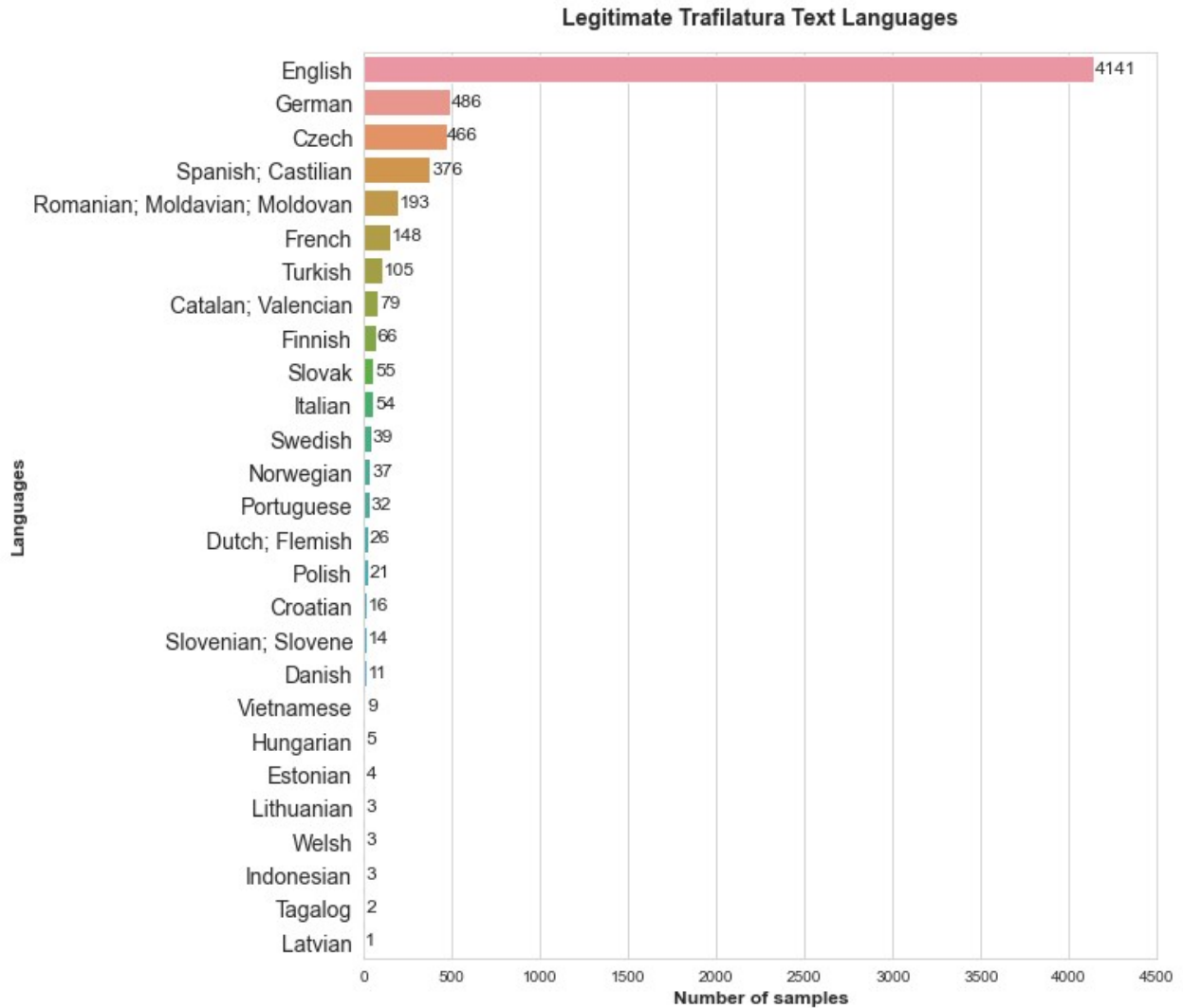
```
plt.title("Legitimate Trafilatura Text Languages", fontsize=14,  
fontweight='bold', y=1.02)  
plt.xlim(0, 4500)
```

Add numbers over the bars

```
for p in plot.patches:  
    plot.annotate(format(int(p.get_width()), 'd'), (p.get_width()+7,  
p.get_y()+0.5), fontsize=12)
```

Show the plot

```
plt.subplots_adjust(left=0.2, right=0.9, top=0.9, bottom=0.1) # You  
may need to fine-tune these values  
plt.show()
```



```
BeautifulSoup_langs_legit =
legit_samples.BeautifulSoup_text_language.value_counts()
```

```
# top 10 languages detected in BeautifulSoup text
BeautifulSoup_langs_legit[:10]
```

```
English          4151
German           479
Czech            463
Spanish; Castilian 385
Romanian; Moldavian; Moldovan 199
French           147
Turkish          106
Catalan; Valencian 77
Finnish           66
Slovak           56
Name: BeautifulSoup_text_language, dtype: int64
```

```

print('The total number of legitimate languages (BS): ',
len(BeautifulSoup_langs_legit))

The total number of legitimate languages (BS): 27

langs = BeautifulSoup_langs_legit.index
values = BeautifulSoup_langs_legit.values

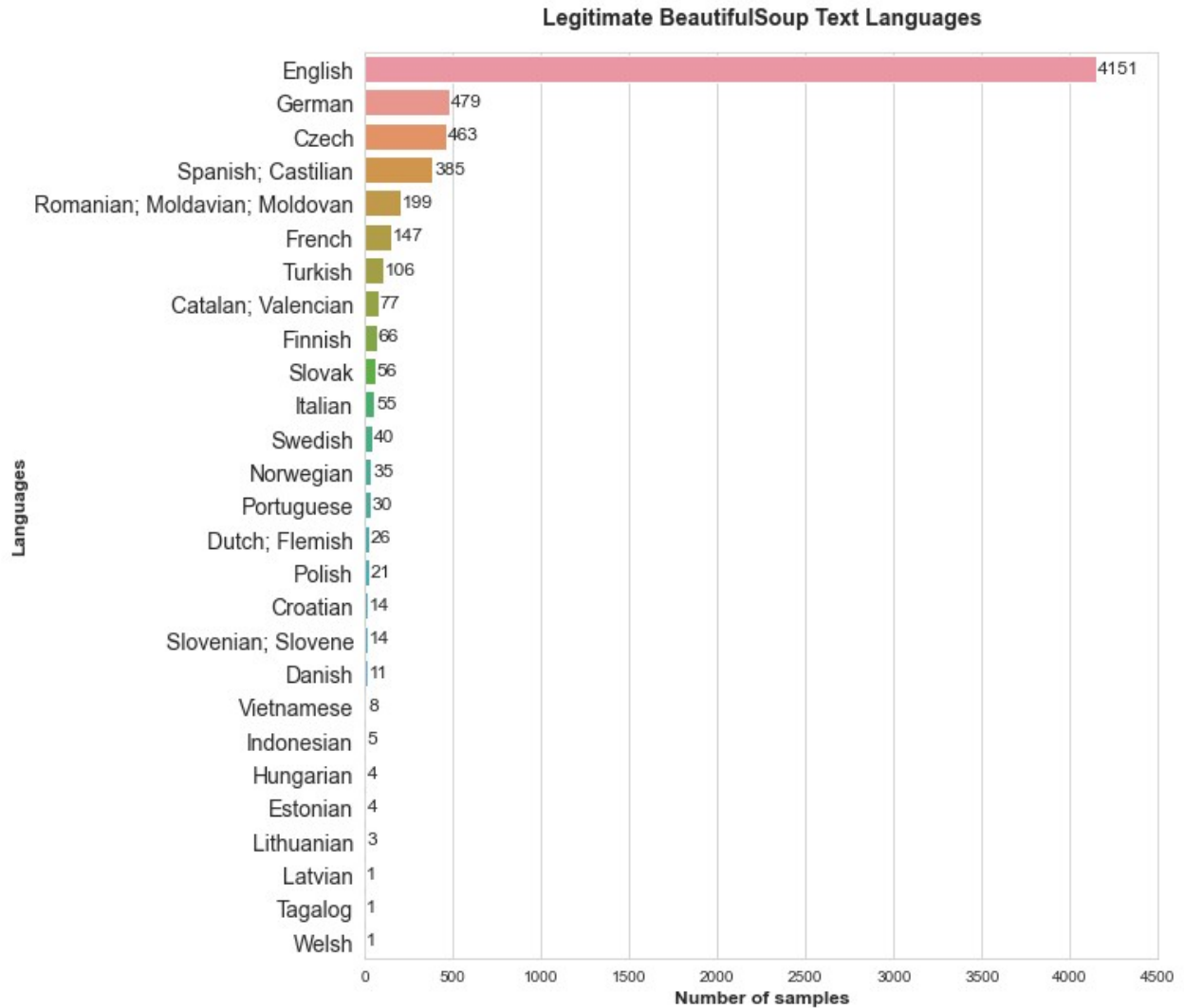
# Create the plot
sns.set_style("whitegrid")
plt.figure(figsize=(10, 10)) # Adjust the figure size accordingly
plot = sns.barplot(x=values, y=langs, orient='h')

# Increase the font size of y-axis labels
plot.set_yticklabels(plot.get_yticklabels(), fontsize=14)

plt.xlabel('Number of samples', fontsize=12, fontweight='bold')
plt.ylabel('Languages', fontsize=12, fontweight='bold')

# plt.title("Top 10 Languages in Legit Trafilatura Text")
plt.title("Legitimate BeautifulSoup Text Languages", fontsize=14,
fontweight='bold', y=1.02)
plt.xlim(0, 4500)
# Add numbers over the bars
for p in plot.patches:
    plot.annotate(format(int(p.get_width()), 'd'), (p.get_width()+7,
p.get_y()+0.5), fontsize=12)
# Show the plot
plt.subplots_adjust(left=0.2, right=0.9, top=0.9, bottom=0.1) # You
may need to fine-tune these values
plt.show()

```

```
html_langs_legit = legit_samples.html_text_language.value_counts()
```

```
# top 10 languages detected in html text
```

```
html_langs_legit[:10]
```

```
English          6211
Czech            50
Catalan; Valencian  36
French           34
Spanish; Castilian  21
Romanian; Moldavian; Moldovan  19
German           13
Slovak           6
Italian           5
Danish           5
Name: html_text_language, dtype: int64
```

```

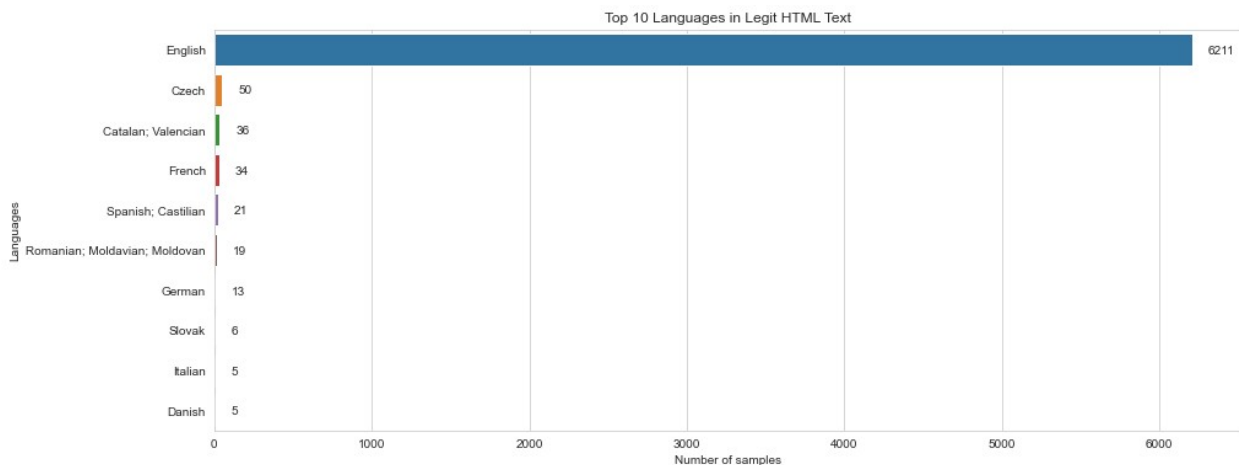
langs = html_langs_legit[:10].index
values = html_langs_legit[:10].values

# Create the plot
sns.set_style("whitegrid")
plt.figure(figsize=(15,6))
plot = sns.barplot(x=values, y=langs, orient='h')

# Add labels and title
plt.xlabel('Number of samples')
plt.ylabel('Languages')
plt.title("Top 10 Languages in Legit HTML Text")
# Add numbers over the bars
for p in plot.patches:
    plot.annotate(format(int(p.get_width()), 'd'), (p.get_width()+100,
p.get_y()+0.5))

# Show the plot
plt.show()

```



2.6 SSL Distribution

```

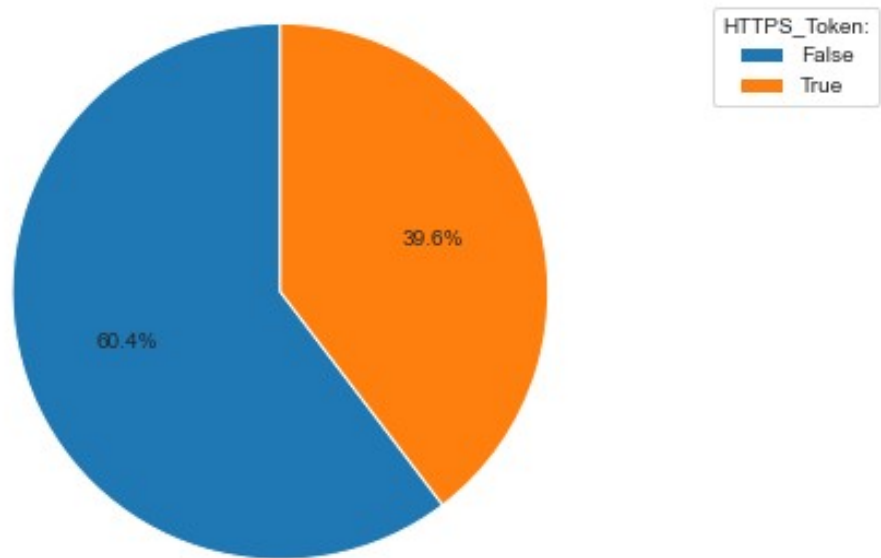
legit_samples.SSL.value_counts()

False    3873
True      2543
Name: SSL, dtype: int64

sizes = legit_samples.SSL.value_counts().values
labels= legit_samples.SSL.value_counts().index
fig=plt.figure(figsize=(10, 5))
plt.pie(sizes, startangle = 90, autopct = '%1.1f%%')
plt.legend(labels, title = 'HTTPS_Token:')
plt.axis('equal')

```

```
plt.show()
```



2.7 Top 10 TLD & Domains

```
legit_samples.Domain.value_counts()[ :10]
```

```
wikipedia      23
makeuseof      10
ehow            10
computerhope   10
streamingmedia  6
businessdictionary  6
howtogeek      6
pinterest      5
explainthatstuff  5
livestrong     5
Name: Domain, dtype: int64
```

```
domains = legit_samples.Domain.value_counts()[ :15].index
counts = legit_samples.Domain.value_counts()[ :15].values
```

```
# Set the figure size
plt.figure(figsize=(15, 6))
```

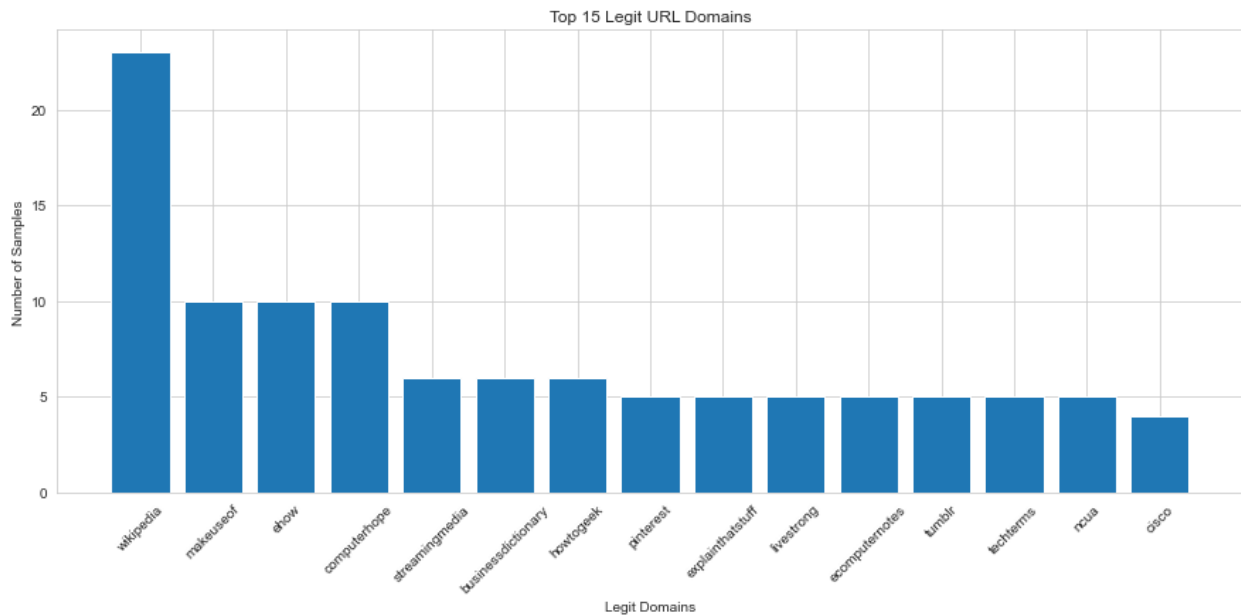
```
# Create the bar plot
plt.bar(domains, counts)
```

```
# Add labels and title
plt.xlabel("Legit Domains")
plt.ylabel("Number of Samples")
```

```
plt.title("Top 15 Legit URL Domains")

# Rotate the x-tick labels
plt.xticks(rotation=45)

# Show the plot
plt.show()
```



```
legit_samples.TLD.value_counts()[:10]

com      3019
org       610
cz        446
de        371
ro        188
es        144
net       144
edu        82
fi         81
co.uk     77
Name: TLD, dtype: int64

tlds = legit_samples.TLD.value_counts()[:15].index
counts = legit_samples.TLD.value_counts()[:15].values

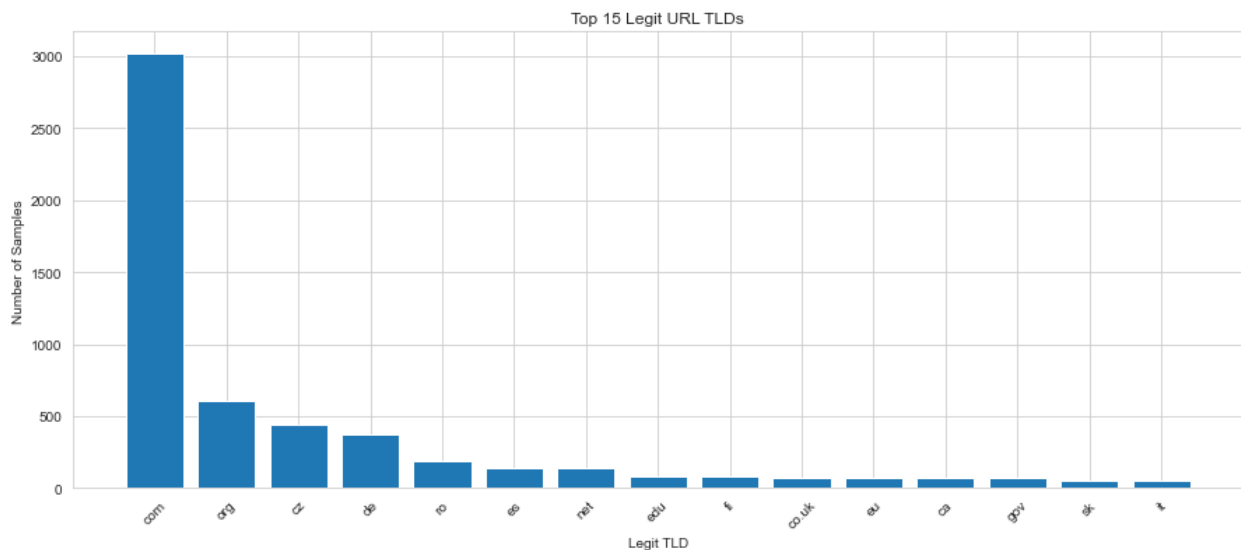
# Set the figure size
plt.figure(figsize=(15, 6))

# Create the bar plot
plt.bar(tlds, counts)
```

```
# Add labels and title
plt.xlabel("Legit TLD")
plt.ylabel("Number of Samples")
plt.title("Top 15 Legit URL TLDs")

# Rotate the x-tick labels
plt.xticks(rotation=45)

# Show the plot
plt.show()
```



2.8 URL Length Stats

```
print('Legit URL average length:',
      round(legit_samples.URL.str.len().sum()/sum(legit_samples.URL.notnull()), 4))
print('Legit URL min length:', legit_samples.URL.str.len().min())
print('Legit URL max length:', legit_samples.URL.str.len().max())
print('Legit URL median length:',
      legit_samples.URL.str.len().median())
```

```
Legit URL average length: 44.3003
Legit URL min length: 14
Legit URL max length: 342
Legit URL median length: 35.0
```

2.9 Top 10 FLD & Subdomains

```
legit samples.iloc[:, 4:9].head(10)
```

	URL	TLD
Domain \		

```

0      https://www.depicus.com/wake-on-lan/\n      com
depicus
1      http://liquidsdr.org/doc/tutorial-fec/\n      org
liquidsdr
2      http://www.csgnetwork.com/ctrcalc.html\n      com
csgnetwork
3      http://www.dailymusicsheets.com/piano/sheets/\n      com
dailymusicsheets
4      http://www.mylifetime.com/shows/the-client-list\n      com
mylifetime
5      http://www.gpsinstallations.co.uk/services/aud...  co.uk
gpsinstallations
6      http://www.echoecho.com/html/links01.htm\n      com
echoecho
7      https://plato.stanford.edu/entries/voltaire/\n      edu
stanford
8      http://www.123helpme.com/tps-a-transaction-pro...  com
123helpme
9      https://www.brandsoftheworld.com/logo/save-the...  com
brandsoftheworld

```

	FLD	Subdomain
0	depicus.com	www
1	liquidsdr.org	None
2	csgnetwork.com	www
3	dailymusicsheets.com	www
4	mylifetime.com	www
5	gpsinstallations.co.uk	www
6	echoecho.com	www
7	stanford.edu	plato
8	123helpme.com	www
9	brandsoftheworld.com	www

```

print('the number of unique fld:', len(legit_samples.FLD.unique()))
legit_samples.FLD.value_counts()[:20]

```

```

the number of unique fld: 5705

```

wikipedia.org	23
makeuseof.com	10
computerhope.com	10
ehow.com	8
howtogeek.com	6
businessdictionary.com	6
streamingmedia.com	6
pinterest.com	5
livestrong.com	5
ncua.gov	5
techterms.com	5
explainthatstuff.com	5

```
ecomputernotes.com      5
tumblr.com              5
ussoccerda.com          4
sapo.mz                 4
ghdp.com                4
msubulldogclub.com     4
graphic-design-employment.com 4
ravensburger-gruppe.de 4
Name: FLD, dtype: int64
```

```
print('the number of unique subdomains:',
len(legit_samples.Subdomain.unique()))
legit_samples.Subdomain.value_counts()[:20]
```

```
the number of unique subdomains: 468
```

```
www      4487
en        32
ru        11
shop      10
www.cs    10
blog       8
wiki       7
app        7
web        5
docs       5
taxi       4
news       4
extranet   4
source     4
m          4
forum      4
corp       4
online     4
en.m       4
pro        3
```

```
Name: Subdomain, dtype: int64
```

```
legit_samples.iloc[:, 4:9].describe()
```

	URL	TLD
Domain \		
count	6416	6408
6408		
unique	6378	196
5692		
top	http://bestreviews.com/best-wireless-routers\n	com
wikipedia		
freq	2	3019
23		

	FLD	Subdomain
count	6408	5131
unique	5704	467
top	wikipedia.org	www
freq	23	4487

Key Findings & Quality Assessment

Uniqueness Analysis

- **URL Uniqueness:** High/Low (Refer to 1.8)
- **Content Uniqueness:** High/Low (Refer to describes)

Observation: Compare the `unique` counts in the `df.describe()` outputs above. A significant drop from URL unique count to Content (HTML/Text) unique count indicates data redundancy.

Multimodal Integrity

- Check for non-null values in `trafilatura_text`, `BeautifulSoup_text`, and `image_path` columns in Section 1.1 and 2.1.